Exercise #2 - Performance analysis of G1

Wind Tunnel Testing of Wind Turbines MW2285

Lehrstuhl für Windenergie

Problem definition

In this exercise you are going to study the performance of the scaled wind turbine model *G1* (MATLAB or python recommended). The data is included in the `LookUpTable_G1.mat` available in Moodle [1]. The variables are listed below:

- `Pitch`: pitch angles in degrees
- `TSR`: tip-speed ratios
- `cord`: blade chord
- `rm`: spanwise blade location
- `Cp`: power coefficient C_p for `Pitch` and `TSR`
- `Cf`: **thrust** coefficient C_T for `Pitch` and `TSR`
- `Ct`: torque coefficient C_Q for `Pitch` and `TSR`
- `Axial`: axial induction coefficient a for `Pitch` and `TSR`
- `Alpha`: angle of attack for `Pitch` and `TSR`



Figure 1: G1 wind turbine model.

Table 1: Specifics of the G1 wind turbine model.

Diameter, D	1.1 m
Rated wind speed, U_r	5.7 m/s
Rated rotational speed, ω_r	850 RPM
Rated tip speed ratio (TSR), λ_r	8.58 [-]
Rated power, P_r	46.02 W
Hub height	0.825 m
Tip Speed	48.96 m/s
Rated torque, Q_r	0.511 Nm
Reynolds number	85,000 [-]

Tasks

1. Plot the chord distribution and compute the Reynolds number at the mid span of the blade. Assume rated ω and an optimal value for the axial induction $a = 1/3$ [-].
   Hint: remember the velocity triangle from Assignment 1.
2. For a wind turbine, the power coefficient C_p indicates the efficiency of the rotor in terms of energy extraction from the incoming wind. Using the data provided, to determine the highest obtainable power coefficient C_p, and the corresponding values of the tip speed ratio and pitch angle, λ_{opt} , θ_{opt}.

[1] The data can be loaded with `load` in MATLAB, or with `scipy.io.loadmat` in python.

3. Based on the data provided, what effect does pitching has on the C_p? What about rotor acceleration/deceleration (i.e. different λ)?
   Hint #1: move from the optimal point found in Question 2 and observe how C_p changes.
   Hint #2: $\lambda = \frac{\omega R}{U_\infty}$

4. Repeat, this time with C_T.

5. Repeat the analysis of Task 2 for the other Reynolds numbers available in the table. How does the maximum C_p changes with the Reynolds number?

6. Plot the distributions of axial induction, angle of attack. At the blade extremities, you should clearly see the effects of root/tip vortices.